Customer Shopping Behaviour Analysis

Analysing 3,900 transactions to identify patterns in customer spending, segmentation, product preferences, and subscription behaviour.

Dataset Overview

3,900

Transactions

Total records analysed

18

Features

Data points per transaction

37

Missing Values

Missing ratings imputed

Customer Attributes

- Age, Gender, Location
- Subscription Status

Transaction Details

- Item, Category, Amount
- Season, Size, Colour

Behavioural Metrics

- Discounts, Promo Codes
- Frequency, Ratings

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Data Preparation Process

01

Data Ingestion

Loaded dataset using pandas

02

Initial Analysis

Evaluated structure with .info() and .describe()

03

Missing Value Treatment

Imputed missing ratings using category-wise median

04

Feature Engineering

Created age groups and purchase frequency metrics

05

Validation & Export

Removed redundancy, exported to PostgreSQL

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Revenue by Gender

Key Insight

SQL queries compared total spending across male and female customers to identify gender-based purchasing patterns.

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High-Value Discount Users

Target Segment

Customers using discounts but spending above average

Strategy

Identify high-value customers who respond to promotions

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Product Performance Analysis

Top-Rated Products

Products ranked by average review ratings

Discount Dependency

Products heavily reliant on discounts for sales

Top Products by Category

Best-performing products within each category

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Shipping & Subscription Impact

Shipping Comparison

Spending patterns compared between Standard and Express shipping options.

Subscription Analysis

Revenue and average spend evaluated for subscribers versus non-subscribers.

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Customer Segmentation

New

First-time customers

Returning

Multiple purchases

Loyal

High-frequency buyers

Repeat purchase behaviour analysed for correlation with subscription likelihood.

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Revenue by Age Group

Age Segmentation

Customers classified into age groups using binned ranges

Contribution Analysis

Revenue contribution assessed for each age segment

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Business Recommendations

Increase Subscription Adoption

Offer exclusive incentives to convert regular customers into subscribers

Strengthen Loyalty Programs

Reward repeat customers to improve retention and lifetime value

Optimize Discount Strategy

Balance promotional impact with profitability

Enhance Product Visibility

Promote high-performing and top-rated products

Targeted Marketing

Focus campaigns on high-value segments and express-shipping users

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